Agent-guided model exploration for CYP inhibition prediction

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Abstract

We used research agents to compare molecular representations, auxiliary assays, architectures, and ensembles for CYP inhibition prediction. An audited sequence of 540 scored evaluations covered 433 within-campaign recipe groups and represented 1,543 model fits. A hit-expansion evaluation approximated the challenge's selection of analogs around potent training compounds. Improvements on this split did not consistently transfer to the leaderboard. The final candidate combines 40 Chemprop networks. This study illustrates how agents can broaden and document experimental exploration, while also exposing the limits of adaptive internal evaluation.

1 Research process and scope

Humans set the objectives and constraints; agents maintained an idea queue, implemented experiments, interpreted results, and revised the queue. Scripts dispatched training jobs and monitored execution. This separation allowed long-running searches without continuous model interaction. Negative results informed subsequent experiments rather than disappearing from the record.

The audit counts completed scored evaluations, including controls and repeated initialization seeds (Fig. 1). Recipe groups are deduplicated within campaigns, not across the entire project. Fit counts expand each evaluation's folds or rotations. Failed or incomplete jobs, smoke tests, duplicate records, submission refits, and calibration-only processing are excluded. Separate foundation-model and specialist studies fall outside this total. These numbers measure exploration, not independent discoveries. We did not compare productivity against a human-only research team.

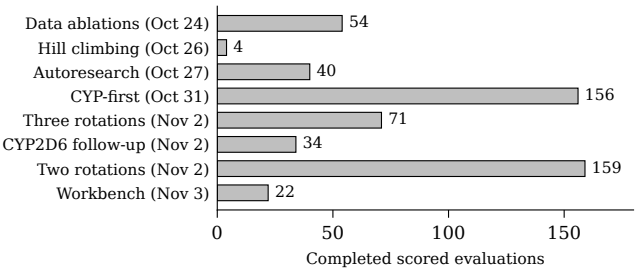


Figure 1: Completed scored evaluations by campaign. The final 22 configurations account for 440 model fits; evaluations therefore differ in computational cost.

2 Molecular representations

Morgan fingerprints encode fixed local two-dimensional substructures and provide inexpensive inputs for tree and linear controls. Chemprop models learn atom–bond message-passing representations from assay labels, with shared encoders supporting auxiliary outputs. CheMeleon [1] adds a descriptor-pretrained graph encoder; comparisons included frozen features, tree and neural heads, and encoder adaptation. A SMILES transformer instead learns from molecular strings. Its completed pilot did not outperform its graph reference.

Atom-JEPA [2] tested a different source of information: atom types and three-dimensional coordinates. We generated conformers with RDKit ETKDGv3/MMFF94 and compared 256-dimensional scalar features with 768-dimensional features incorporating higher-order invariant norms. Further experiments varied scaling, heads, task weights, losses, and conformer sampling. Rotation and translation checks, together with checkpoint replay, tested numerical consistency. Multi-conformer training and prediction averaging were treated separately; an averaging gain alone does not establish a benefit from geometry. Atom-JEPA improved one CYP1A2 development comparison but did not show a consistent advantage across evaluations. Mol-JEPA is retained in the historical record but excluded from the corporate-use candidate set. Neither JEPA model contributes to the final predictor.

3 Evaluation under hit expansion

The official challenge expanded the top 25 training hits for CYP1A2, CYP2C9, and CYP3A4 into close analogs; CYP2D6 did not nominate hits [3]. Our analogous split reserved half of standardized official parents as a training seed pool (random seed 42). Only seed-pool labels selected hits, using $\mathrm{pIC}_{50} \geq 5$ for the three nominating CYPs. Structures alone selected up to ten analogs per hit from the other half, using 2,048-bit Morgan fingerprints (radius 2) and Tanimoto similarity ≥ 0.50. We used a larger potent-hit bank than the official top-25 quota to improve coverage.

Overlapping nomination families remained together when balanced into groups A and B; neither assay values nor assay availability determined assignment. All bank and official blind parents were excluded from every training source, including standardized aliases. Potent seed hits remained eligible for training. Training stayed fixed while A and B exchanged checkpoint-validation and scoring roles (Fig. 2). Each scored compound therefore used a fresh fit whose checkpoint was selected on the other group.

The bank contains only 42, 38, 42, and 156 measured labels for CYP1A2, CYP2C9, CYP2D6, and CYP3A4, respectively. Reducing three rotations to two enlarges each scoring group without adding observations. Comparisons use the same saved split and matched initialization seeds; scores from the three-rotation split are not directly compared. Repeated adaptive use makes these development results, not independent test estimates.

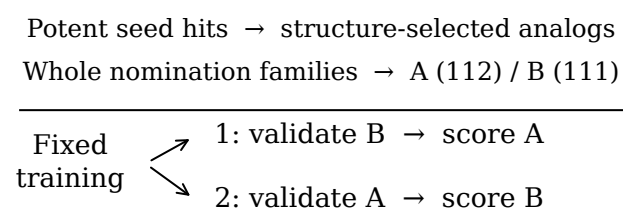


Figure 2: Two-rotation evaluation. The 223-parent analog bank is excluded from the fixed training set of 32,686 records. Validation selects checkpoints; scoring predictions are pooled across rotations.

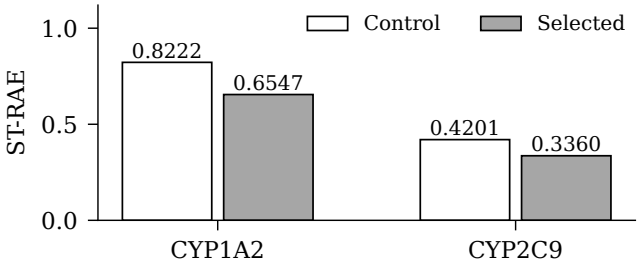


Figure 3: Matched hit-expansion development results after averaging raw predictions over three seeds. Lower ST-RAE is better. Both selected variants improved error in both rotations; CYP2D6 and CYP3A4 controls were retained.

The CYP1A2 variant combined Atom-JEPA, a Huber head, task weighting, a lower learning rate, and removal of qHTS auxiliaries. The CYP2C9 variant removed TDI auxiliaries. CYP1A2 subsequently worsened on Workbench and the official leaderboard, motivating a return to Workbench-led selection. Structural resemblance did not guarantee transfer. The final ensemble has not been freshly evaluated on the analog bank.

4 Final ensemble

Official challenge, ChEMBL, and Veith/PubChem measurements are represented as separate masked outputs [4, 5]. Two freshly trained Chemprop configurations each use four task bundles and five preserved Workbench scaffold folds, giving 40 networks. The two configurations use depth 5/7, width 700, dropout 0.10/0.05, auxiliary weight 0.30/0.60, and peak learning rate 0.001/0.0005. Both use two width-300 head layers, weighted MAE, batch size 64, training-only target scaling, and early stopping (400 epochs maximum; patience 40). Blind and fold-validation parents are excluded from training. Predictions are averaged equally over folds, applicable bundles, and configurations in raw pIC_{50} units.

Matched Workbench macro out-of-fold (OOF) ST-RAE improved from 0.63943 to 0.61159 (4.35%), with lower error and higher Spearman correlation for all four endpoints. These checkpoint- and ensemble-selected results are development evidence, distinct from Fig. 3.

Endpoint	Control	Final ensemble
CYP1A2	0.75764	0.72200
CYP2C9	0.55774	0.51686
CYP2D6	0.82446	0.81078
CYP3A4	0.41786	0.39672

After raw averaging, predictions were rescaled using the historical Workbench leaderboard-informed transform [4]. The accompanying reproducibility note records the exact postprocessing and its provenance.

The candidate uses no proprietary training data or external prediction files. Its 750-row CSV passed local validation; no official score for this candidate is claimed. Code, report source, and aggregate figure data are provided in the accompanying repository; trained checkpoints are not included.

References

[1] CheMeleon. arXiv:2506.15792.

[2] Atom-JEPA. Molecular checkpoint and model card.

[3] OpenADMET. CYP challenge design and interim results, 2026.

[4] ADMET Workbench. CYP challenge working notes.

[5] OpenADMET. CYP challenge training and test dataset.